

17 March 2025

Monday

Polynomial Regression

Given data:

Experience (x)	Salary (y)
1	40000
2	45000
3	53000
4	67000
5	85000
6	109000
7	140000

Step 1: Form the Design Matrix

For a quadratic polynomial ($n=2$), the equation is

$$Y = b_0 + b_1 X + b_2 X^2$$

we construct a matrix with polynomial features:

$$X_{\text{poly}} = \begin{bmatrix} 1 & X_1 & X_1^2 \\ 1 & X_2 & X_2^2 \\ 1 & X_3 & X_3^2 \\ \vdots & \vdots & \vdots \\ 1 & X_7 & X_7^2 \end{bmatrix}$$

Example For $X = 1, 2, 3, 4, 5, 6, 7$:

$$X_{\text{poly}} = \begin{bmatrix} 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \\ 1 & 4 & 4^2 \\ 1 & 5 & 5^2 \\ 1 & 6 & 6^2 \\ 1 & 7 & 7^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \end{bmatrix}$$

Step 2: Create the target vector Y .

$$Y = \begin{bmatrix} 40000 \\ 45000 \\ 53000 \\ 67000 \\ 85000 \\ 109000 \\ 140000 \end{bmatrix}$$

~~Step 3: Create the target vector Y~~

Step 3: Compute B using the Normal Equation

X^T ← Transpose

$$\cancel{B = (X^T X)^{-1} X^T Y} \quad B = (X^T X)^{-1} X^T Y$$

→ First, compute $X^T X$:

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 4 & 9 & 16 & 25 & 36 & 49 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 4 & 9 & 16 & 25 & 36 & 49 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \end{bmatrix} = \begin{bmatrix} 7 & 28 & 140 \\ 28 & 140 & 784 \\ 140 & 784 & 4634 \end{bmatrix}$$

Step

Inverse of Matrix $(X^T X)^{-1}$

$$A = X^T X = \begin{bmatrix} 7 & 28 & 140 \\ 28 & 140 & 784 \\ 140 & 784 & 4634 \end{bmatrix}$$

Step 1: Compute the Determinant $|A|$

The determinant of a 3×3 matrix:

$$|A| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Expanding:

$$|A| = 7 \begin{vmatrix} 140 & 784 \\ 784 & 4634 \end{vmatrix} - 28 \begin{vmatrix} 28 & 784 \\ 140 & 4634 \end{vmatrix} + 140 \begin{vmatrix} 28 & 140 \\ 140 & 784 \end{vmatrix}$$

To solve these, we calculate 2×2 determinants:

$$\begin{vmatrix} 140 & 784 \\ 784 & 4634 \end{vmatrix} = (140 \times 4634) - (784 \times 784) = 648760 - 614656 = 34104$$

$$\begin{vmatrix} 28 & 784 \\ 140 & 4634 \end{vmatrix} = (28 \times 4634) - (784 \times 140) = 129752 - 109760 = 19992$$

$$\begin{vmatrix} 28 & 140 \\ 140 & 784 \end{vmatrix} = (28 \times 784) - (140 \times 140) = 21952 - 19600 = 2352$$

Now, substitute back:

$$|A| = (7 \times 34104) - (28 \times 19992) + (140 \times 2352)$$

$$|A| = 8232$$

$$A = \begin{vmatrix} 7 & 28 & 140 \\ 28 & 140 & 784 \\ 140 & 784 & 4634 \end{vmatrix}$$

Step 2: Compute the Cofactor Matrix C

The cofactor of element a_{ij} is computed by removing row i and column j ; then computing the determinant of the remaining 2×2 matrix.

Each cofactor:

$$C_{11} = \begin{vmatrix} 140 & 784 \\ 784 & 4634 \end{vmatrix} = 34104$$

$$C_{12} = - \begin{vmatrix} 28 & 784 \\ 140 & 4634 \end{vmatrix} = -19992$$

$$C_{13} = \begin{vmatrix} 28 & 140 \\ 140 & 784 \end{vmatrix} = 2352$$

$$C_{21} = - \begin{vmatrix} 28 & 140 \\ 784 & 4634 \end{vmatrix} = -19992$$

$$C_{22} = \begin{vmatrix} 7 & 140 \\ 140 & 4634 \end{vmatrix} = 12838$$

$$C_{23} = - \begin{vmatrix} 7 & 28 \\ 140 & 784 \end{vmatrix} = -1568$$

$$C_{31} = \begin{vmatrix} 28 & 140 \\ 140 & 784 \end{vmatrix} = 2352$$

$$C_{32} = - \begin{vmatrix} 7 & 140 \\ 28 & 784 \end{vmatrix} = -1568$$

$$C_{33} = \begin{vmatrix} 7 & 28 \\ 28 & 140 \end{vmatrix} = 196$$

$$C = \begin{bmatrix} 34104 & -19992 & 2352 \\ -19992 & 12838 & -1568 \\ 2352 & -1568 & 196 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 34104 & -19992 & 2352 \\ -19992 & 12838 & -1568 \\ 2352 & -1568 & 196 \end{bmatrix}$$

Step 4: Compute the Inverse $A^{-1} = \frac{1}{|A|} C^T$

$$A^{-1} = \begin{bmatrix} \frac{34104}{8232} & \frac{-19992}{8232} & \frac{2352}{8232} \\ \frac{-19992}{8232} & \frac{12838}{8232} & \frac{-1568}{8232} \\ \frac{2352}{8232} & \frac{-1568}{8232} & \frac{196}{8232} \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 4.143 & -2.429 & 0.286 \\ -2.429 & 1.560 & -0.190 \\ 0.286 & -0.190 & 0.024 \end{bmatrix}$$

Compute $X^T Y$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 4 & 9 & 16 & 25 & 36 & 49 \end{bmatrix} \begin{bmatrix} 40000 \\ 45000 \\ 53000 \\ 67000 \\ 85000 \\ 109000 \\ 140000 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 40000 + 45000 + 53000 + 67000 + 85000 + 109000 + 140000 \\ 40000 + 90000 + 159000 + 268000 + 425000 + 654000 + 980000 \\ 40000 + 180000 + 477000 + 1072000 + 2125000 + 3924000 + 6860000 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 539000 \\ 2616000 \\ 14678000 \end{bmatrix}$$

Multiply $(X^T X)^{-1}$ and $X^T Y$

$$B_0 = (4.143 \times 539000) + (-2.429 \times 2616000) + (0.286 \times 14678000) \\ = 2233077 - 6354264 + 4197908$$

$$B_0 = 76721$$

$$B_1 = (-2.429 \times 539000) + (1.560 \times 2616000) + (-0.190 \times 14678000) \\ = -1309231 + 4080960 - 2788820$$

$$B_1 = -17091$$

$$B_2 = (0.286 \times 539000) + (-0.190 \times 2616000) + (0.024 \times 14678000) \\ = 154154 - 497040 + 352272 \\ = 9386$$

$$B = \begin{bmatrix} 76721 \\ -17091 \\ 9386 \end{bmatrix}$$

$$\text{Salary} = 76721 - 17091x + 9386x^2$$

